



## 1. Background

Rwanda and many African countries are multilingual societies where communication happens across several languages including Kinyarwanda, English, and French. Despite this linguistic diversity, access to fast, affordable translation services remains limited. Individuals in transportation, education, tourism, research, and daily life often struggle to communicate effectively.

Traditional translation solutions are slow, expensive, or require trained human professionals. Digital tools like Google Translate offer basic support but lack real-time translation accuracy for Kinyarwanda, speech-to-text transcription, or automatic video translation—creating significant gaps for local users.

There is a growing need for an accessible, AI-driven solution that supports both written and spoken translation tailored to local languages, with offline functionality for areas with limited internet.

## 2. Problem Statement

Language barriers remain a major challenge in Rwanda and across Africa. Many people interact daily across languages—English, French, and Kinyarwanda—but lack fast, affordable translation tools.

For example:

- Tourists struggle to communicate with locals.
- Motorcyclists (moto riders), Tax-men, some cannot understand passengers speaking foreign languages, leading to confusion and safety risks.
- Parents struggle to follow school communications not written in Kinyarwanda.
- Students cannot fully access online educational content because most videos, lectures, and research materials are in foreign languages.

Existing translation services are slow, costly, or require human translators. Current digital tools do not support real-time Kinyarwanda translation, accurate speech-to-text, or automatic translation of speeches and YouTube videos into Kinyarwanda.

This leaves millions without access to essential information, creating communication gaps, learning obstacles, and lost opportunities in daily life.

## 3. Project Solution

We are developing a mobile and web application powered by Artificial Intelligence that instantly translates speech, text, and online videos into Kinyarwanda and other major languages.



PRIMEDATA CONSULTING GROUP LTD

Kigali-Rwanda

Email: [primedataconsultinggroup.rw@gmail.com](mailto:primedataconsultinggroup.rw@gmail.com)

## Key Features

1. Video Translation
  - a. Translate YouTube and other online videos into Kinyarwanda
  - b. Automatic subtitles and transcripts
  - c. Voice-over translation options
2. Real-Time Speech Translation
  - a. Speak any language, receive instant audio and text translation
  - b. Ideal for riders, tourists, teachers, and service providers
3. Motorist Mode (Hands)
  - a. Designed for motorcyclists and drivers
  - b. Audio-first translation for safe communication while moving
  - c. Map with clear routes, road signs, and voice guidance.
4. Camera Translation
  - a. Audio books, school notes, or documents
  - b. Translate the text into local or other languages
5. Offline Translation Packs
  - a. Translation without active internet
  - b. Useful in rural areas and low-connectivity environments
6. Education Tools
  - a. Auto-summarization of lectures
  - b. Study notes generation
  - c. Vocabulary extraction

## Target Users

- Tourists and visitors
- Motorcyclists and drivers

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- Researchers and journalists
- Parents, Students
- Teachers and lecturers
- Public service providers
- General public in Rwanda and Africa

#### **4. Project Mission**

To break language barriers by providing fast, reliable, and accessible translation tools that empower people in Rwanda and Africa to communicate, learn, and connect with the world.

#### **5. Project Vision**

To become Africa's leading language-technology platform making information, education, and communication accessible to everyone, regardless of the language they speak.

#### **6. Objectives**

##### **6.1 Overall Objective**

To create a mobile and web application translation platform that enhances communication, supports education, and improves access to information across African languages.

##### **6.2 Specific Objectives**

- Provide real-time speech translation in Kinyarwanda and major global languages.
- Enable automatic translation of online content, particularly educational video materials.
- Improve road communication and safety through Motorist Mode.
- Support schools, universities, and families with accurate translation of academic content.
- Offer offline and low-data translation options for underserved communities.
- Facilitate tourism and international business communication in Rwanda.

#### **7. Expected Outcomes**

- Improved communication between people speaking different languages.
- Increased access to education, especially multilingual online content.
- Better support for tourists and international visitors.
- Reduced cost and time spent on manual translation.



- Enhanced safety and service quality for motorcyclists and public transport providers.
- Increased digital inclusion for rural and low-income communities.
- Growth of local language technology innovation in Rwanda.

## 8. Innovation and Uniqueness

The project introduces multiple innovations not currently available in Rwanda:

- AI-powered real-time Kinyarwanda or Other language translation
- Automatic Video and Audio-to-Kinyarwanda or Other language subtitle generation
- Motorist Mode, optimized for hands-free use
- Offline-capable translation technology
- Integrated education tools for students and teachers
- A localized approach that respects dialects, accents, and cultural context

This combination of AI speech, translation, and offline accessibility makes the platform one of the first of its kind in the region.

## 9. Implementation Plan

### Phase 1: Research & Prototype Development

- Build core AI translation and speech modules
- Create basic mobile app interface
- Languages of interest: Kinyarwanda, English, and French

### Phase 2: Pilot Testing

- Test with motorcyclists, tourists
- Refine accuracy for Kinyarwanda and regional languages

### Phase 3: Full App Development

- Add offline features
- Add camera translation
- Expand to multiple languages

### Phase 4: Launch & Scale-Up



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- Release on Android, iOS, and web
- Partner with schools, tourism agencies, moto cooperatives, and NGOs
- Expand to neighboring countries

## 10. Sustainability / Revenue Model

Even with a free version, income can be generated through:

- Premium subscriptions (advanced features)
- School and university partnerships
- NGO and government contracts
- Translation marketplace commissions
- API access for businesses
- Light advertising
- Pay-per-use for long video translations

This ensures long-term sustainability and growth.

## 11. Conclusion

This project addresses one of the most persistent barriers in African societies—cross-language communication. By leveraging AI, the platform makes translation faster, cheaper, and more accessible to millions of people. It supports education, tourism, transportation, and general communication while empowering Rwanda as a leader in language technology innovation.

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Email: [primedataconsultinggroup.rw@gmail.com](mailto:primedataconsultinggroup.rw@gmail.com)

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